

Kapitel 1.3

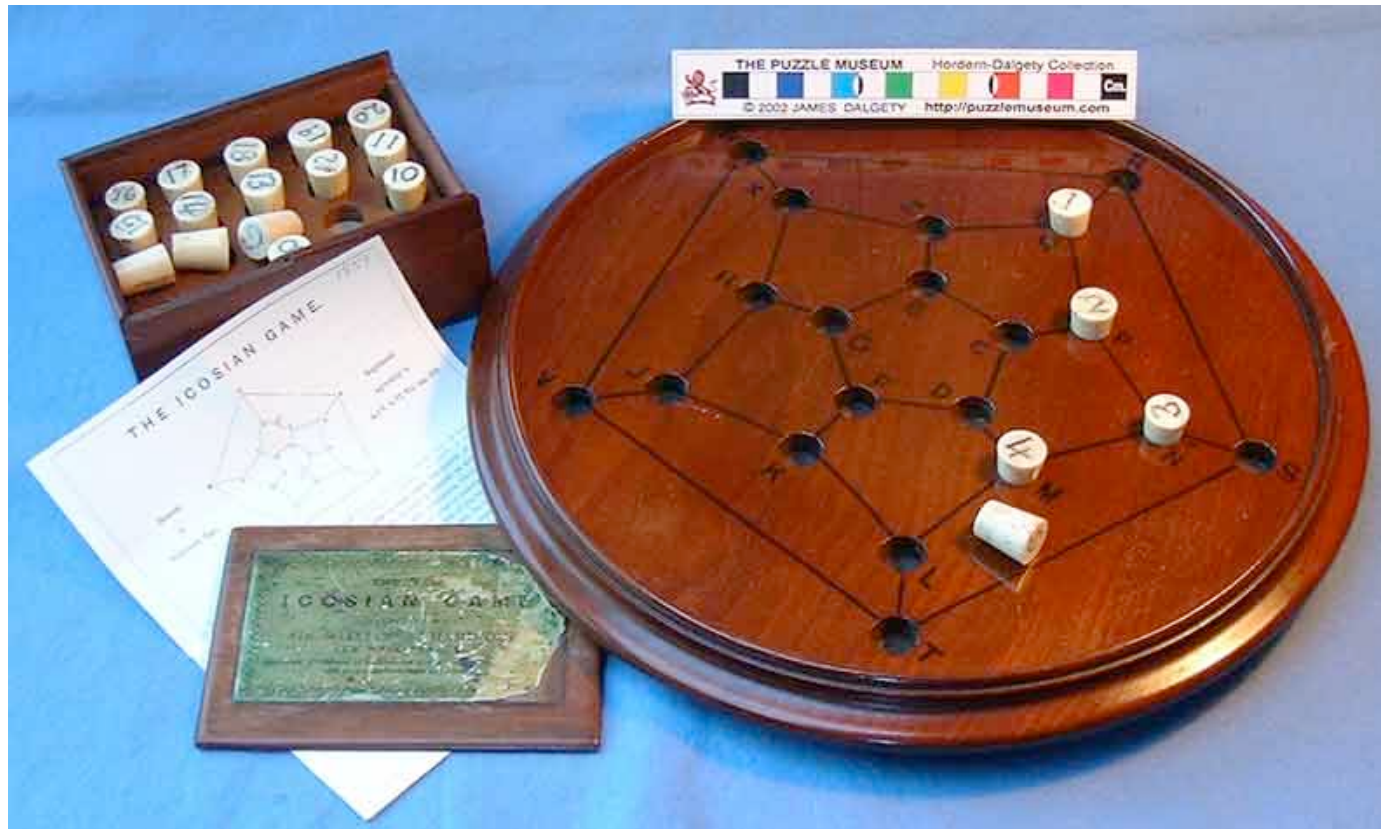
Kreise

- Sei $G = (V, E)$ ein Graph.
- **Eulertour:**
 - Ein geschlossener Weg in G , der jede **Kante** genau einmal enthält.
- **Hamiltonkreis:**
 - Ein Kreis in G , der jeden **Knoten** genau einmal enthält.

Satz: Ein zusammenhängender Graph $G = (V, E)$ enthält eine Eulertour

gdw. der Grad jedes Knotens gerade ist.

... und eine solche kann man in $O(|E|)$ Zeit finden



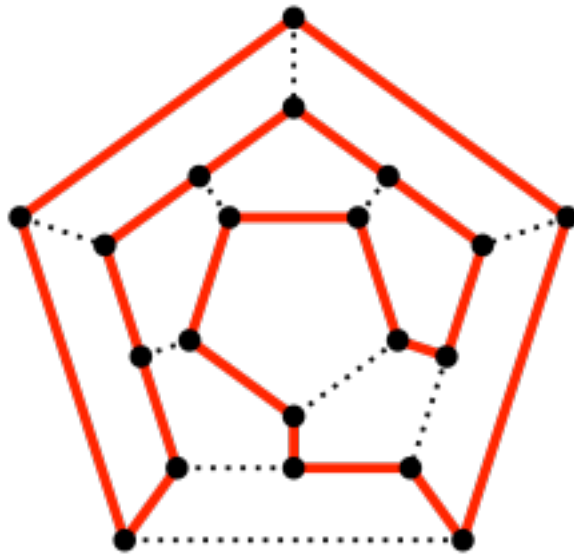
Ikosaeder Spiel



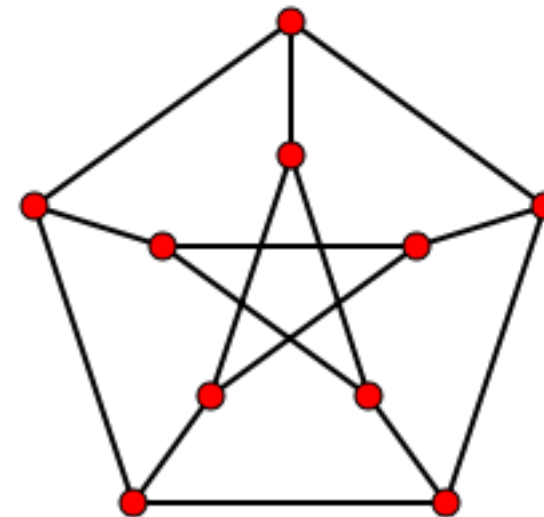
Sir William Hamilton
(1805 - 1865)

Gesucht: Hamiltonkreis



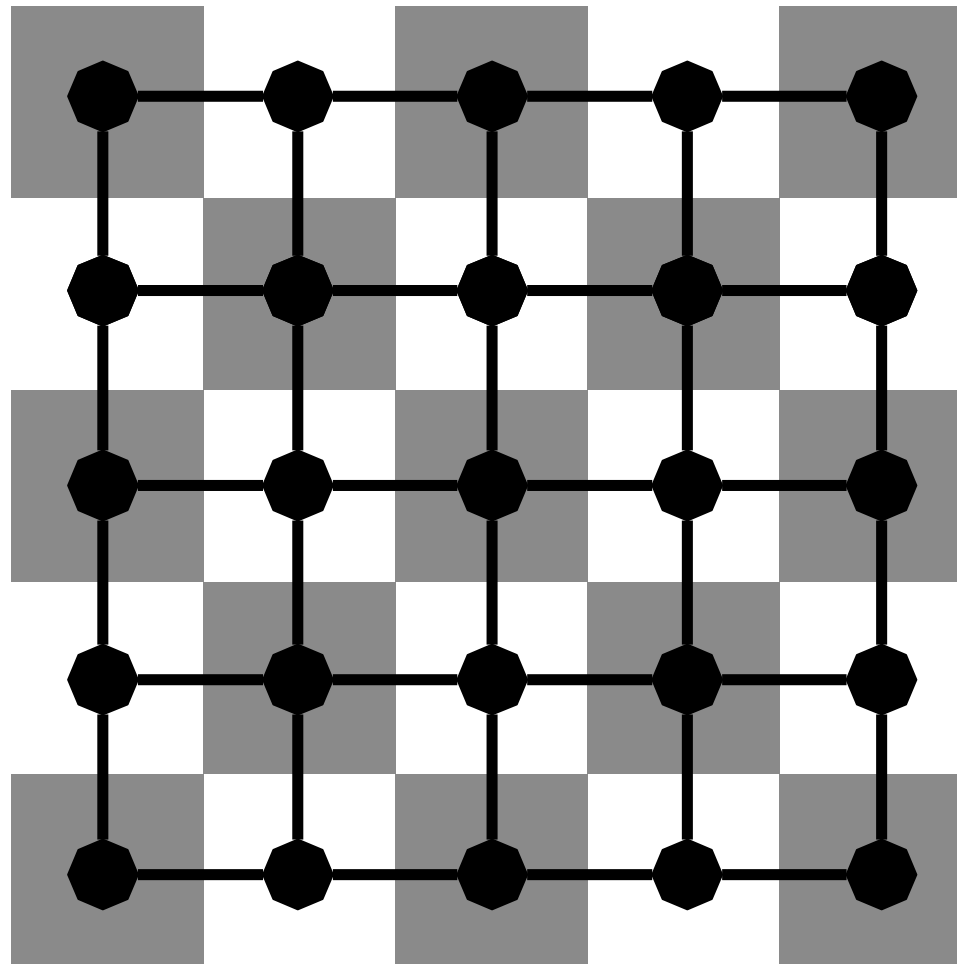


Ikosaeder



Petersengraph





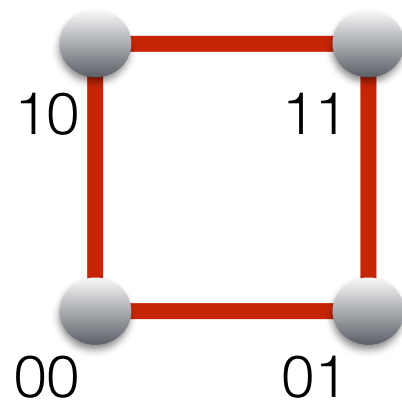
Satz: Seien $m, n \geq 2$.

Ein $n \times m$ Gitter enthält einen Hamiltonkreis gdw $n \cdot m$ gerade ist.

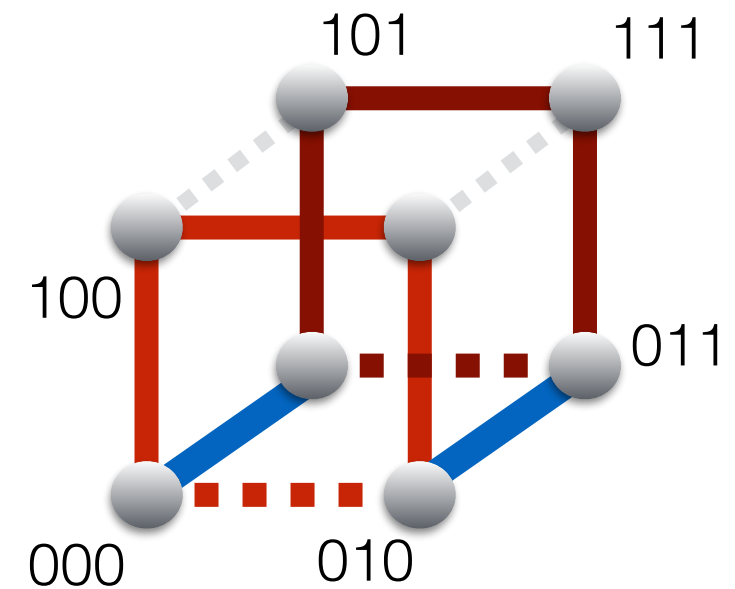
Satz: Ein bipartiter Graph $G = (A \cup B, E)$ mit $|A| \neq |B|$ kann keinen Hamiltonkreis enthalten.

Hamiltonkreis im Hyperwürfel: Gray-Code

d=2:



d=3:



00
10
11
01

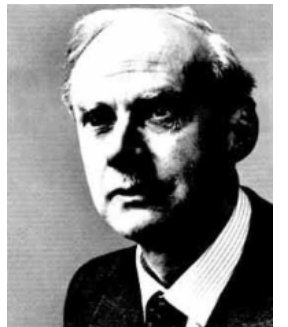
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analog für $d \geq 4$

Satz: (Dirac 1952)

Jeder Graph $G = (V, E)$ mit $|V| \geq 3$ und Minimalgrad

$\delta(G) \geq |V|/2$ enthält einen Hamiltonkreis.

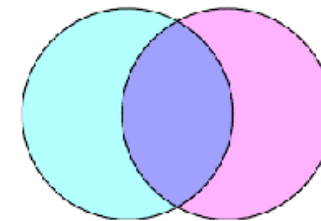


Paul Dirac
(1902-1984)

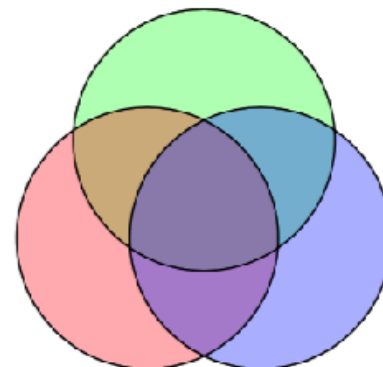
Satz 1.35. (*Siebformel, Prinzip der Inklusion/Exklusion*) Für endliche Mengen $A_1, \dots, A_n$ ($n \geq 2$) gilt:

$$\begin{aligned} \left| \bigcup_{i=1}^n A_i \right| &= \sum_{l=1}^n (-1)^{l+1} \sum_{1 \leq i_1 < \dots < i_l \leq n} |A_{i_1} \cap \dots \cap A_{i_l}| \\ &= \sum_{i=1}^n |A_i| - \sum_{1 \leq i_1 < i_2 \leq n} |A_{i_1} \cap A_{i_2}| + \sum_{1 \leq i_1 < i_2 < i_3 \leq n} |A_{i_1} \cap A_{i_2} \cap A_{i_3}| \\ &\quad - \dots + (-1)^{n+1} \cdot |A_1 \cap \dots \cap A_n|. \end{aligned}$$

Für $n=2$: $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|.$



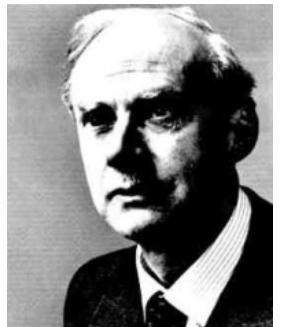
Für $n=3$: $|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3|.$



Satz: (Dirac 1952)

Jeder Graph $G = (V, E)$ mit $|V| \geq 3$ und Minimalgrad

$\delta(G) \geq |V|/2$ enthält einen Hamiltonkreis.



Paul Dirac
(1902-1984)

Wie entscheidet man (effizient),
ob ein Graph einen Hamiltonkreis enthält?

Brute Force: $\approx n!$

Ziel für heute: $\approx 2^n$

Zum Vergleich:

$$\log_2(n!) = \Theta(n \log n)$$
$$\log_2(2^n) = n$$

Wie entscheidet man (effizient),
ob ein Graph einen Hamiltonkreis enthält?

Brute Force: $\approx n!$

Ziel für heute: $\approx 2^n$



For now, general purpose AI models that were trained using
a total computing power of more than 10^{25} FLOPs
are considered to carry systemic risks

$$10^{25} \approx$$

$$25! \approx 1.6 \text{ e}25$$

$$2^{84} \approx 1.9 \text{ e}25$$

Für alle $S \subseteq [n]$ mit $1 \in S$ und alle $x \in S$ mit $x \neq 1$:

„Idee“

$$P_{S,x} := \begin{cases} 1, & \text{es gibt einen 1-x-Pfad, der genau die Knoten aus } S \text{ enthält} \\ 0, & \text{sonst} \end{cases}$$

Initialisierung:

Dynamische Programmierung:

Laufzeit: $\approx n^2 2^n$

Speicherplatz: $\approx n 2^n$

$\forall x \in \{2, \dots, n\}$: für $S = \{1, x\}$ setze $P_{S,x} = 1$ gdw. $\{1, x\} \in E$

Schleife:

for all $s = 3$ to n

for all $S \subseteq [n]$ mit $1 \in S$ und $|S| = s$:

for all $x \in S$ mit $x \neq 1$:

Rekursion

$$P_{S,x} = \max \{ P_{S \setminus \{x\}, y} : y \in N(x) \cap S, y \neq 1 \}$$

Ausgabe: G enthält Hamiltonkreis gdw $\exists x \in N(1)$ mit $P_{[n],x} = 1$

Dynamic Programming: How to brand your research?

The 1950s were not good years for mathematical research.
[The Secretary of Defense] had a pathological fear and hatred of the word “research”.

What title, what name, could I choose? [...] I wanted to get across the idea that this was dynamic, this was multistage, this was time-varying.
[...] It also has a very interesting property as an adjective, and that is it's impossible to use the word dynamic in a pejorative sense.

— Richard Bellman, *Eye of the Hurricane: An Autobiography* (1984)

Dynamic Programming:

How to brand your research?

According to Dr. Darby Saxbe, a professor at the University of Southern California, the following words are considered forbidden or banned and will prompt a review at NSF.

- activism
- activists
- advocacy
- advocate
- barrier
- barriers
- biased
- bias
- BIPOC
- Black and Latinx
- community diversity
- community equity
- cultural differences
- cultural heritage
- culturally responsive
- disabilities
- discrimination
- discriminatory
- backgrounds
- groups
- diversified
- diversify
- enhancing
- equal opportunity
- equality
- equitable
- ethnicity
- excluded
- female
- fostering
- gender
- hate speech
- Hispanic minority
- historically
- implicit bias
- inclusion
- inclusive
- increase
- indigenous community
- inequalities
- inequities
- institutional
- LGBTQ
- marginalize
- minorities
- multicultural
- polarization
- political
- prejudice
- privileges
- promoting
- race
- racial
- justice
- sense of belonging
- sexual preferences
- social justice
- sociocultural
- socioeconomic
- status
- stereotypes
- systemic
- trauma
- underappreciated
- underrepresented
- underserved
- victim
- women

Eulertour vs Hamiltonkreis

Satz: Ein zusammenhängender Graph $G = (V, E)$ enthält eine Eulertour

gdw. der Grad jedes Knotens gerade ist.

... und eine solche kann man in $O(|E|)$ Zeit finden

Satz: Das Problem

„Gegeben ein Graph $G = (V, E)$, enthält G einen Hamiltonkreis?“

kann man in Zeit $O(|V|^2 \cdot 2^{|V|})$ entscheiden und, falls ja, einen solchen finden.

Eulertour vs Hamiltonkreis

Satz: Ein zusammenhängender Graph $G = (V, E)$ enthält eine Eulertour

gdw. der Grad jedes Knotens gerade ist.

... und eine solche kann man in $O(|E|)$ Zeit finden

Satz: (Karp 1972)

Das Problem

„Gegeben ein Graph $G = (V, E)$, enthält G einen Hamiltonkreis?“

ist NP-vollständig.

Exkurs

(Vorgriff auf Vorlesung Theoretische Informatik)

polynomiell

Eulertour

Hamiltonkreis

P = *effizient entscheidbare Probleme*

NP = *(einseitig) effizient verifizierbare Probleme*

nichtdeterministisch polynomiell

$P \stackrel{?}{=} NP$

→ 1 Million US-\$ (Clay-Foundation)

[eines von sieben Millennium-Problemen]

Ein Problem π aus NP heisst **NP -vollständig**, falls gilt

$$\pi \in P \Rightarrow P = NP$$

Es gibt **sehr viele** NP-vollständige Probleme:

- Hamiltonkreis
- Rucksackproblem
- Clique: Gibt es in einem Graphen k paarweise benachbarte Knoten?
- Nullstelle mod n : Hat ein Polynom mod n eine Nullstelle?
- Satisfiability: hat eine logische Formel eine Lösung?
-

... **sehr sehr viele**:

Classic Nintendo Games are
(Computationally) Hard

Greg Aloupis* Erik D. Demaine[†] Alan Guo^{†‡} Giovanni Viglietta[§]

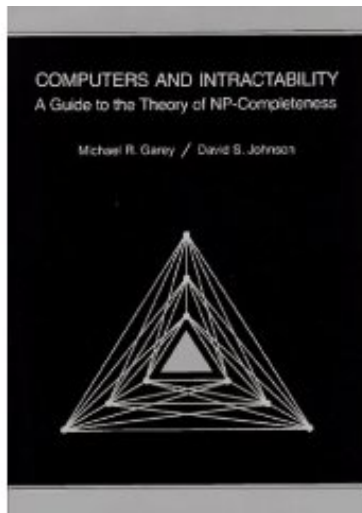
February 10, 2015

Abstract

We prove NP-hardness results for five of Nintendo's largest video game franchises: Mario, Donkey Kong, Legend of Zelda, Metroid, and Pokémon. Our results apply to generalized versions of Super Mario Bros. 1–3, The Lost Levels, and Super Mario World; Donkey Kong Country 1–3; all Legend of Zelda games; all Metroid games; and all Pokémon role-playing games. In addition, we prove PSPACE-completeness of the Donkey Kong Country games and several Legend of Zelda games.

1 Introduction

A series of recent papers have analyzed the computational complexity of playing many different video games [1, 4, 5, 6], but the most well-known classic Nintendo games have yet to be included among these results. In this paper, we analyze some of the best-known Nintendo games of all time: Mario, Donkey Kong, Legend of Zelda, Metroid, and Pokémon. We prove that it is NP-hard, and in some cases PSPACE-hard, to play generalized versions of most games in these series. In



“I can’t find an efficient algorithm, I guess I’m just too dumb.”

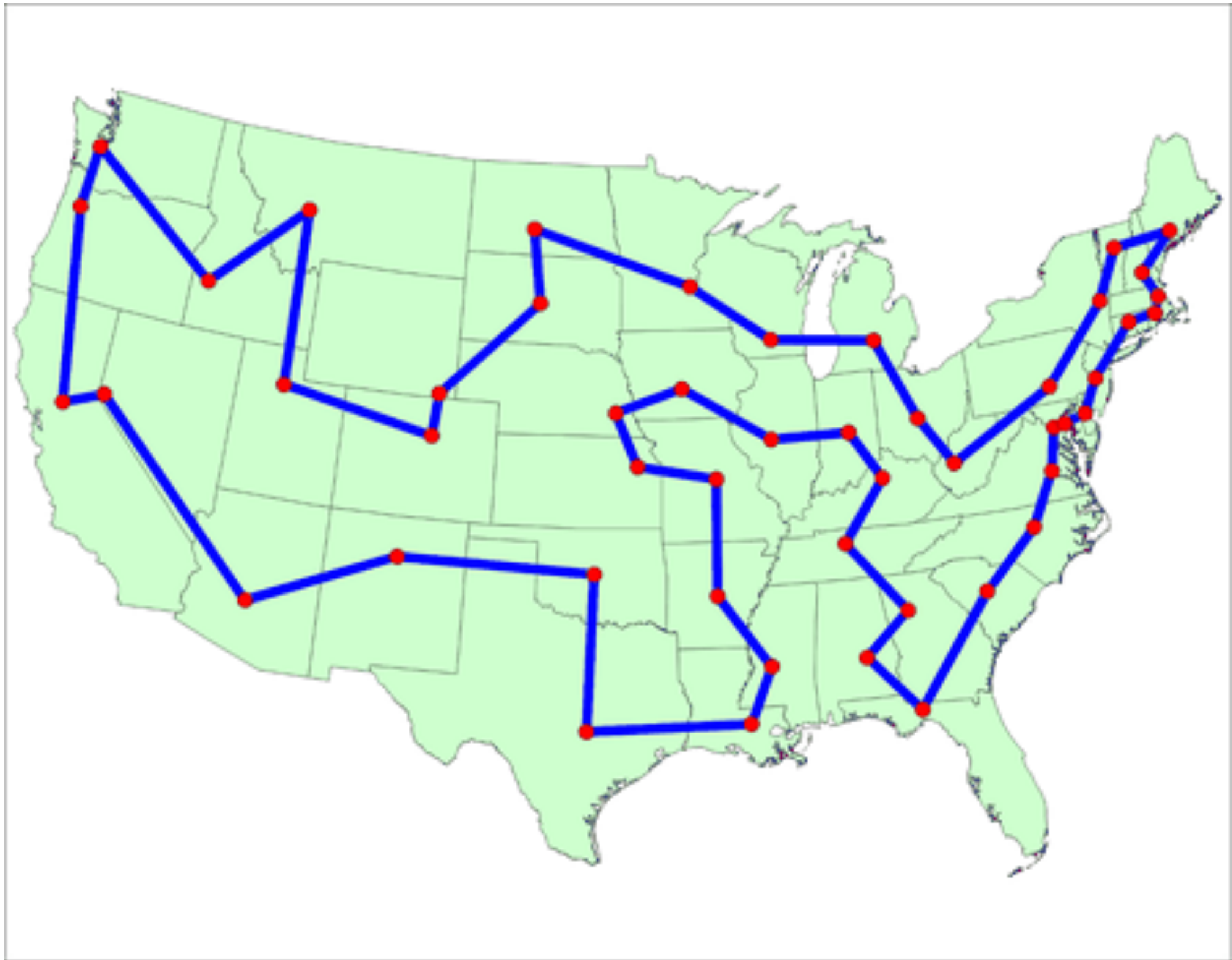


“I can’t find an efficient algorithm, because no such algorithm is possible!”



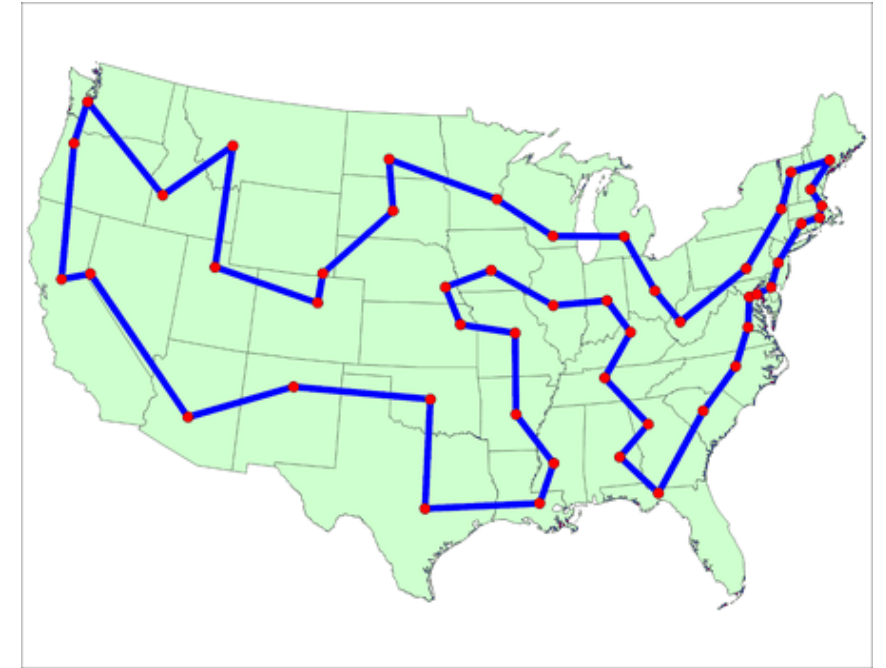
“I can’t find an efficient algorithm, but neither can all these famous people.”

Hamiltonkreis - Beispiele



The Traveling Salesman Problem — Das Problem des Handlungsreisenden

Traveling Salesman Problem



Gegeben: vollständiger Graph auf n Knoten,
Distanzen zw. je zwei Knoten: $\ell : \binom{[n]}{2} \rightarrow \mathbb{R}$

Gesucht? Kürzeste Rundreise:

$$\min_{H: \text{Hamiltonkreis}} \sum_{e \in E(H)} \ell(e)$$

Gesucht: Rundreise kürzer als X (gegeben X)

Traveling Salesman Problem

Das Traveling Salesman Problem (TSP) ist NP-vollständig:

Graph $G = ([n], E)$ $\iff$ vollst Graph $K_n = ([n], \binom{[n]}{2})$
mit Längenfunktion $\ell : \binom{[n]}{2} \rightarrow \{0, 1\}$
so dass $\ell(\{x, y\}) = \begin{cases} 0, & \text{falls } \{x, y\} \in E \\ 1, & \text{sonst.} \end{cases}$

Dann gilt:

G enthält Hamiltonkreis $\iff \text{opt}(K_n, \ell) = 0$

Hieraus folgt auch: für kein $C > 0$ kann es einen polynominellen C -Approximationsalgorithmus geben (ausser es gilt $P=NP$).

List of NP-complete problems

🌐 3 languages ▾

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*This is a **dynamic list** and may never be able to satisfy particular standards for completeness. You can help by **adding missing items** with *reliable sources*.*

This is a list of some of the more commonly known problems that are **NP-complete** when expressed as **decision problems**. As there are hundreds of such problems known, this list is in no way comprehensive. Many problems of this type can be found in [Garey & Johnson \(1979\)](#).

Graphs and hypergraphs [edit]

Graphs occur frequently in everyday applications. Examples include biological or social networks, which contain hundreds, thousands and even billions of nodes in some cases (e.g. [Facebook](#) or [LinkedIn](#)).

- 1-planarity^[1]
- 3-dimensional matching^{[2][3]}:SP1
- Bandwidth problem^[3]:GT40
- Bipartite dimension^[3]:GT18
- Capacitated minimum spanning tree^[3]:ND5
- Route inspection problem (also called **Chinese postman problem**) for **mixed graphs** (having both directed and undirected edges). The program is solvable in polynomial time if the graph has all undirected or all directed edges. Variants include the rural postman problem.^[3]:ND25,ND27
- Clique cover problem^{[2][3]}:GT17
- Clique problem^{[2][3]}:GT19
- Complete coloring, a.k.a. achromatic number^[3]:GT5
- Cycle rank
- Degree-constrained spanning tree^[3]:ND1
- Domatic number^[3]:GT3
- Dominating set, a.k.a. domination number^[3]:GT2
 - NP-complete special cases include the **edge dominating set** problem, i.e., the dominating set problem in line graphs. NP-complete variants include the **connected dominating set** problem and the **maximum leaf spanning tree** problem.^[3]:ND2
- Feedback vertex set^{[2][3]}:GT7
- Feedback arc set^{[2][3]}:GT8
- Graph coloring^{[2][3]}:GT4
- Graph homomorphism problem^[3]:GT52
- Graph partition into **subgraphs** of specific types (triangles, **isomorphic subgraphs**, **Hamiltonian** subgraphs, **forests**, **perfect matchings**) are known NP-complete. Partition into **cliques** is the same problem as **coloring** the **complement** of the given graph. A related problem is to find a partition that is optimal terms of the number of edges between parts.^[3]:GT11,GT12,GT13,GT14,GT15,GT16,ND14
- Grundy number of a directed graph.^[3]:GT56
- Hamiltonian completion^[3]:GT34
- Hamiltonian path problem, directed and undirected.^{[2][3]}:GT37,GT38,GT39
- Graph intersection number^[3]:GT59
- Longest path problem^[3]:ND29
- Maximum bipartite subgraph or (especially with weighted edges) **maximum cut**.^{[2][3]}:GT25,ND16
- Maximum common subgraph isomorphism problem^[3]:GT49
- Maximum independent set^[3]:GT20
- Maximum **Induced path**^[3]:GT23
- Minimum maximal independent set a.k.a. minimum independent dominating set^[4]
 - NP-complete special cases include the **minimum maximal matching** problem,^[3]:GT10 which is essentially equal to the **edge dominating set** problem (see above).
- Metric dimension of a graph^[3]:GT61
- Metric k-center
- Minimum degree spanning tree
- Minimum k-cut
- Minimum k-spanning tree
- Steiner tree, or Minimum spanning tree for a subset of the vertices of a graph.^[2] (The minimum spanning tree for an entire graph is solvable in polynomial time.)
- Modularity maximization^[5]
- Monochromatic triangle^[3]:GT6
- Pathwidth,^[6] or, equivalently, **interval thickness**, and **vertex separation number**^[7]
- Rank coloring
- k-Chinese postman
- Shortest total path length spanning tree^[3]:ND3
- Slope number two testing^[8]

Wir kennen Hunderte (Tausende?) von Problemen die NP-vollständig sind!

Deswegen ein sehr wichtiges Werkzeug um zu erklären warum wir für viele Probleme keine effiziente Algorithmen haben



“I can’t find an efficient algorithm, but neither can all these famous people.”

Haskell Program A

```
foldl (+) 0 [1,2,3,4,5]
```

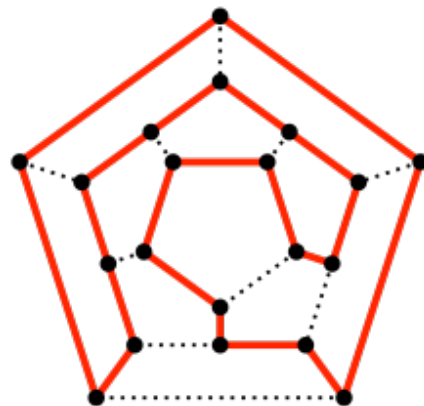
→
A kann als B
kodiert werden

Python Program B

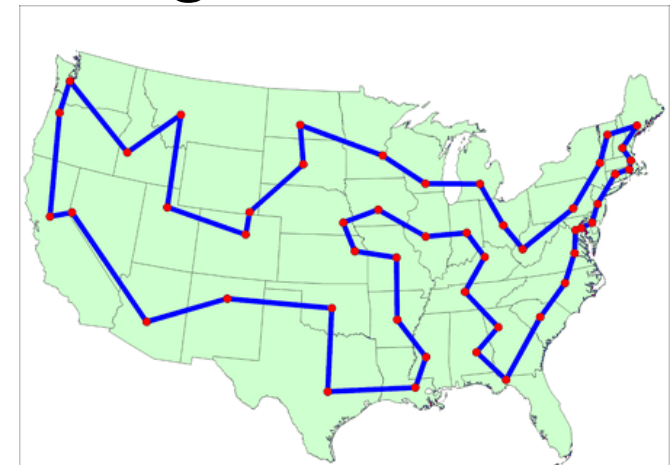
```
from functools import reduce  
xs = [1, 2, 3, 4, 5]  
reduce(lambda x, y: x+y, [1, 2, 3, 4, 5], 0)
```

$$0+1+2+3+4+5 = 15$$

Hamiltonkreis "Program A"



→
TSP "Program B"



Deswegen:

Weil das Hamiltonkreis-Problem NP-vollständig ist,
ist auch das TSP-Problem